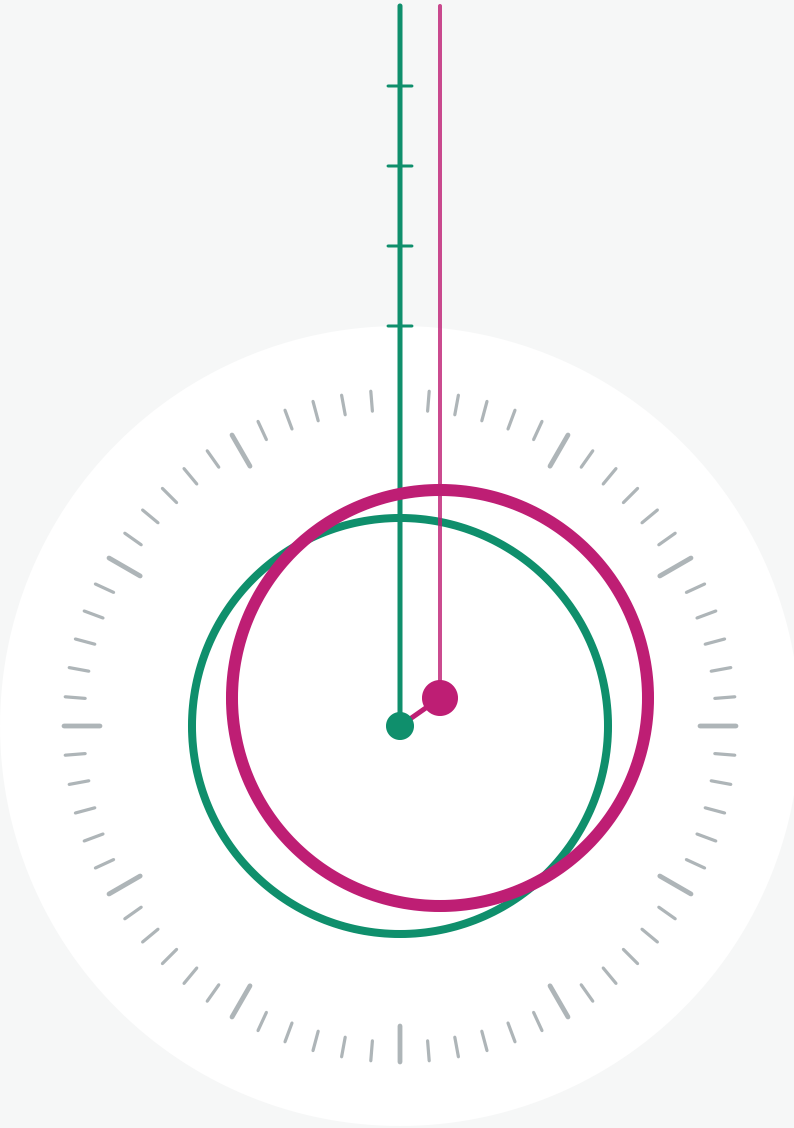
DATUM LINE

Confidence with controls.

Every lab’s cells, on one map.



Kat Titterton, San Francisco



Katherine Titterton

MY WORK

macrofluidic.me/resume

today: search, ask, guess, A25

Do my cells look “normal”?

asked @ five benches
… answered @ none

today: search, ask, guess, A25

One lab, or six?

same cells ✓

same protocol ✓

no drug ✓

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

Could you tell? A classifier can.

99.9% names the right lab, of 11

9% a random guess

fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

One well, onto the map.

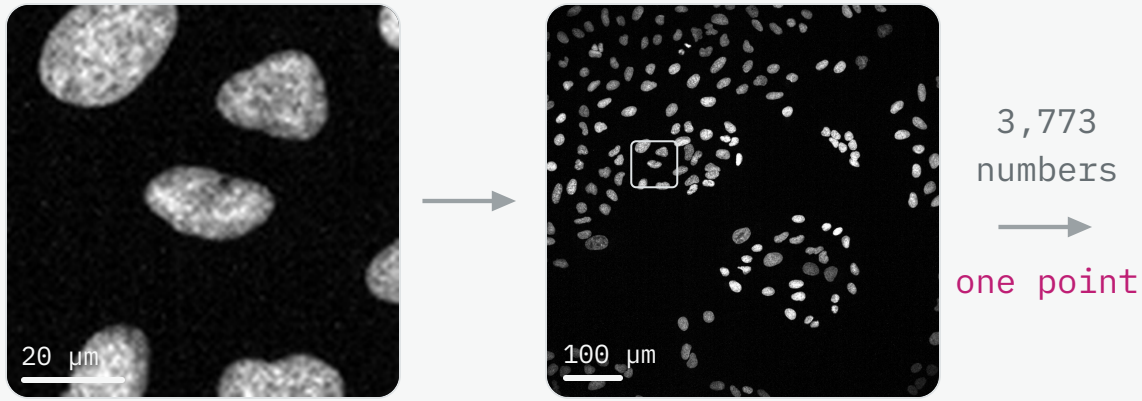
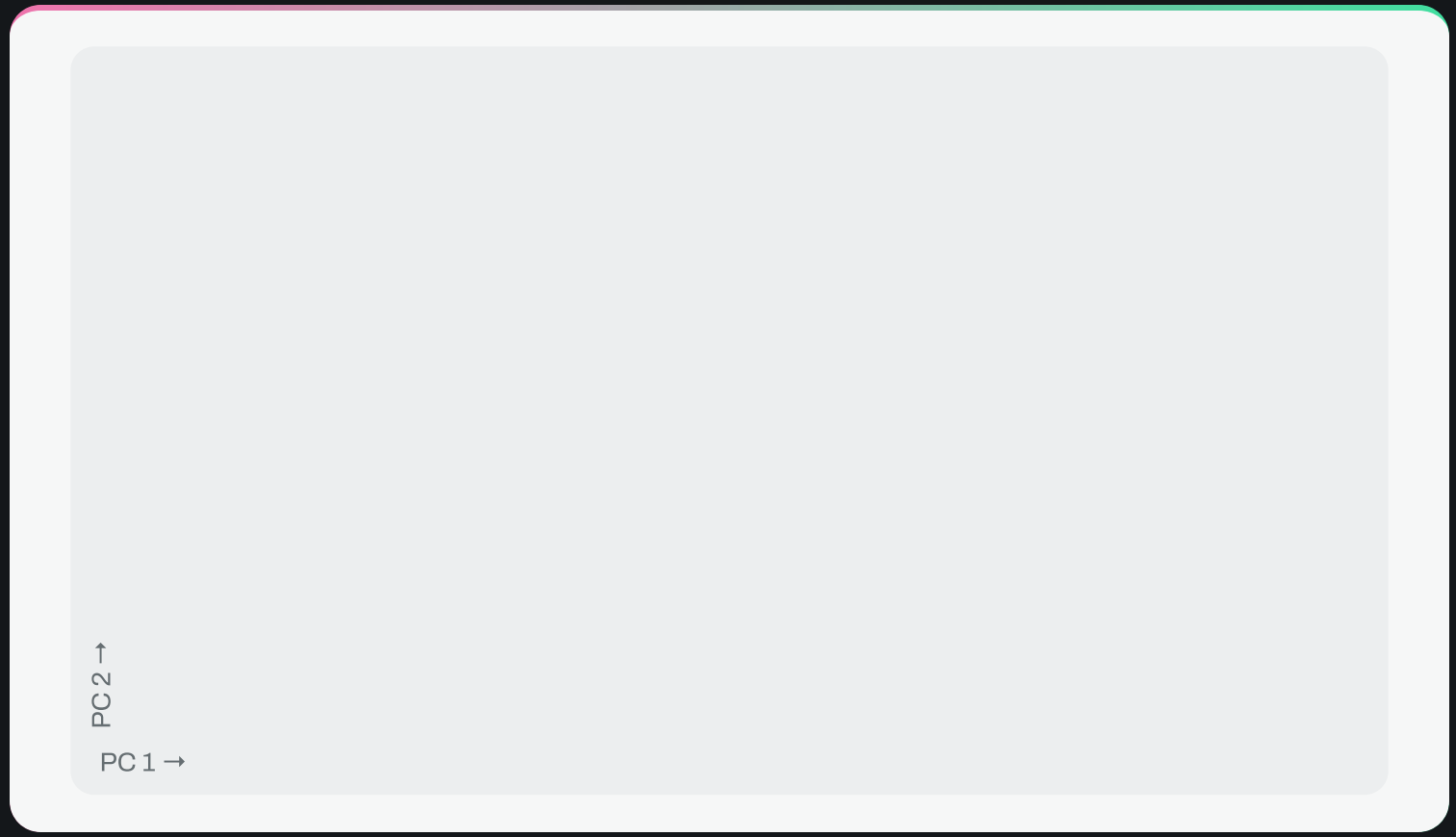




fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

Eleven labs.
Eleven “normals”.



6×

Lab to lab is 6× the spread of wells inside one plate.

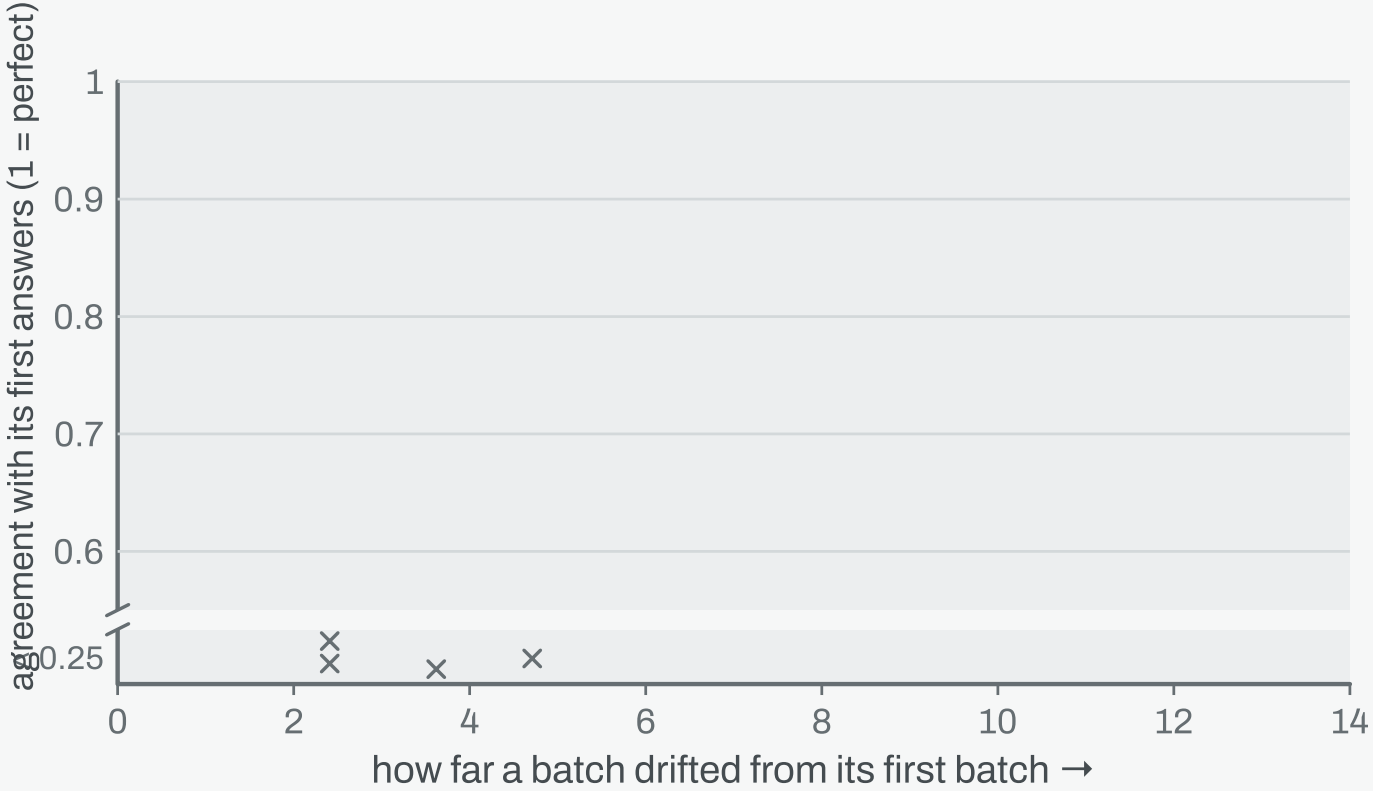
lab to lab

spread inside one plate

6.2 within-plate spreads, lab to lab (results/e13)

fig 2 15,352 DMSO wells, PCA of 3,773 numbers, JUMP [7], A2, A3, A13

Where you sit predicts what you measure.



In every lab: more drift, less agreement. ρ = −0.71 within labs, each lab centred, 118 batches (results/e10)

Lab 2 (you)

From 30 Aug to 3 Oct it drifted 3–5× its summer level.

0.85 → 0.67

Same drugs, checked against its own June answers.

fig 3 118 batches, 11 labs, JUMP [7], A6, A8, A12, A13

Does my control look like yours?

fig 4a RSL3 at JUMP’s stated dose [7], one well per plate, % of cells left = nuclei per field against DMSO wells of the same plate, lab mean, lab 7 (0.625 µM, 80% left) left out, DNA stain, 300 µm, PubMed [16], A9

Does my control look like yours?

fig 4b agreement between pairs of plates, same lab and across labs, 5 µM in every lab [7], 1,829 plates, 11 labs, targets [24], A9, A14

Lab A + lab B ≠ one dataset.

fig 5 EU-OPENSCREEN, one protocol, Hep G2, as reported [13], A23

Every lab that joins sharpens “normal” for everyone.

fig 6 error: distance between two maps built from different labs, in within-plate spreads, measured to five labs, dashed beyond, 1,871 plates, JUMP [7], results/e13

Is my batch “normal”, next to every other lab?

fig 7 replayed on public data, JUMP [7], A11, A18

Which labs make the most reliable data?

fig 8 score = 100 − mean percentile, DMSO wells, JUMP [7], A17

Your certified data, shared on your terms.

fig 9 hollow: off the map, or its known drugs failed (A8), replayed, JUMP [7], terms proposed, A15, A23, A24

Six questions, six tools. Use all of them.

Each answers one question well.
Each misses something, Datum Line too.

Eighteen months, four milestones.

$1.8M

$0.9M line by line, ×2 for the usual overrun A21

→ seed round

Every plate you image, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.

fig 10 1 square = 1 batch, 1,871 plates, JUMP [7], A15, A20, A22, A24

The rest, if you ask.

Every number comes from the code here or a cited source.
Unfinished work is marked pending.

A1 **Why not trust the plate's controls?**
sick plate, sick control

A2 **What is the math?**
six calculations on 3,773 numbers

A3 **Why not batch-correct it away?**
erases most of the lab fingerprint

A4 **Is the model the moat?**
counting cells often wins

A5 **Why not just count cells?**
same count, different texture

A6 **How often do I re-certify?**
your "normal" expires

A7 **Isn't this cell-line DNA testing (STR)?**
which cells, how they behave

A8 **Is the assay itself working?**
known answer

A9 **Does it change the result?**
same drug, different answer

A10 **Does it work without stains?**
bet on brightfield

A11 **Does it flag too often?**
rarely, six of 129 batches

A12 **What moved?**
stain brightness, not cell count

A13 **How does one lab move over time?**
own "normal" fine, other labs: moved

A14 **What does the map buy?**
same drug, closer between labs

A15 **Who pays?**
labs join free, companies pay

A16 **What would the money build?**
ugliest dataset, on purpose

A17 **Is the leaderboard fair?**
ranked on controls, never on results

A18 **What is private?**
control wells leave, nothing else does

A19 **What is the machine learning?**
image in, position and correction out

A20 **What about biosecurity?**
cells show their lab, labels can lie

A21 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A22 **How big?**
count × price, three streams

A23 **What is cell data worth?**
priced in deals, labs disagree

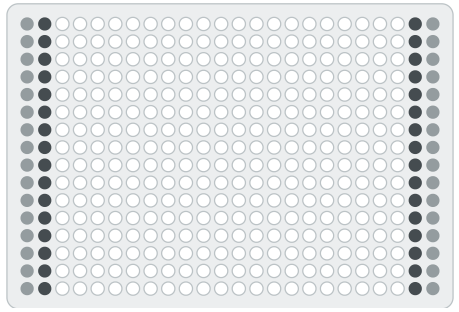
A24 **Why would a lab share?**
owned by the lab, verified by anyone

A25 **How do labs check cells today?**
search, ask, guess, each lab alone

If the plate is sick, its controls are sick too.

Controls sit on the same plate as the tests.

A REAL JUMP PLATE, 384 WELLS



DMSO 32 known effect 32
test 320



test = *"normal"* + *plate* + *effect*
control = *"normal"* + *plate*
difference = *effect on this plate*

plate cancels from the number, never from the cells

WHAT CAN SEE A SICK PLATE?

A view from outside the plate.
Same cells in other labs: the map.

Fig. A1left: JUMP plate 1053601756, lab 2, right: schematic, no measurements

Every result is one of six plain calculations.

Results compare averages, or test a model on unseen plates. Drug discovery removes the lab; a quality check reads it.

The usual correction erases most of the lab's fingerprint.

What counts as "normal" depends on every variable below.
JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

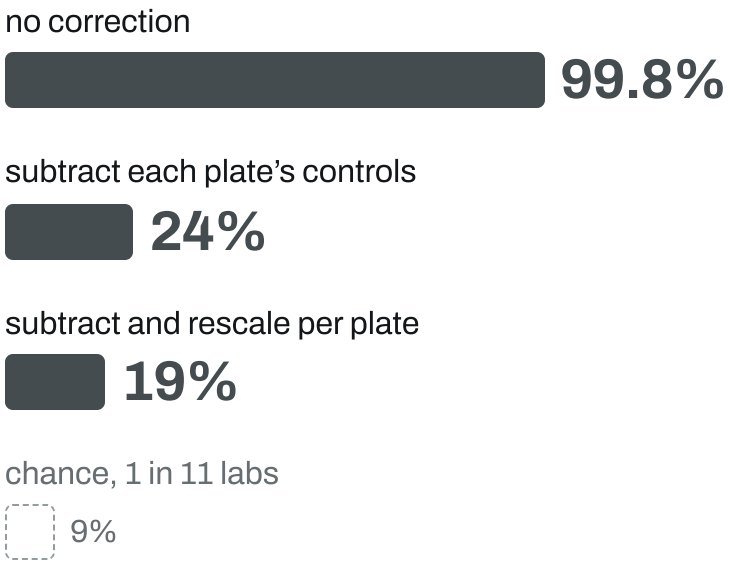
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 DMSO wells, 16 per plate
tested on plates it never saw

Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

On weak benchmarks, counting cells often wins.

I coauthored this paper.
Benchmarks decide which model looks good.

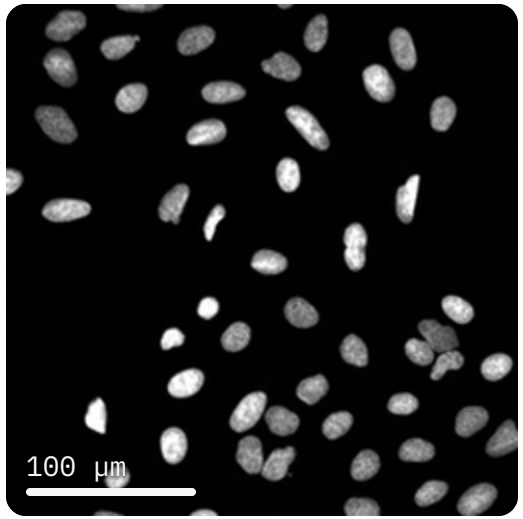
Seal S, Dee W, … **Titterton K**, … Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*, 2026.

Same cell count, different texture: only images see it.

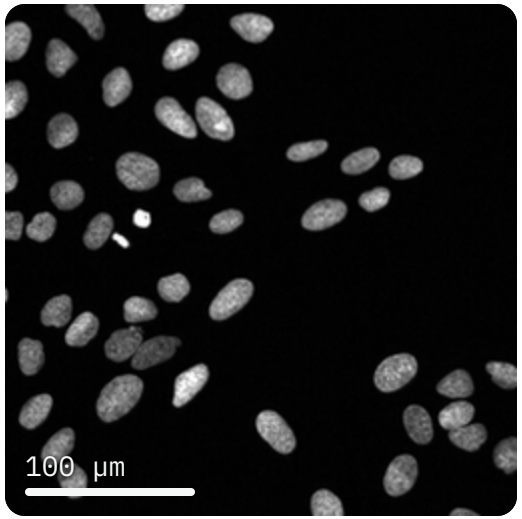
Count moves only when cells die or divide.
Image moves when anything changes.

SAME CELL LINE, SAME PROTOCOL, SAME MICROSCOPE

each lab's typical plate



LAB 5 CV8000, confocal



LAB 6 CV8000, confocal

CELLS COUNTED

196 ≈ 222

gap 0.4× the usual spread → same

FINE DETAIL, % OF DNA SIGNAL

0.94% ≠ 1.10%

gap 2.4× the usual spread → different

fine detail: finer than 5 µm, inside nuclei
usual spread: plate to plate, within one lab

NOT ONE LUCKY PAIR

All 44,325 plate pairs, lab 5 × lab 6:
texture differs in 91%, count in 33%.

TWO WAYS TO READ IT, A2

Classifier: does it belong with past runs?
Distance: how far out, which way back?

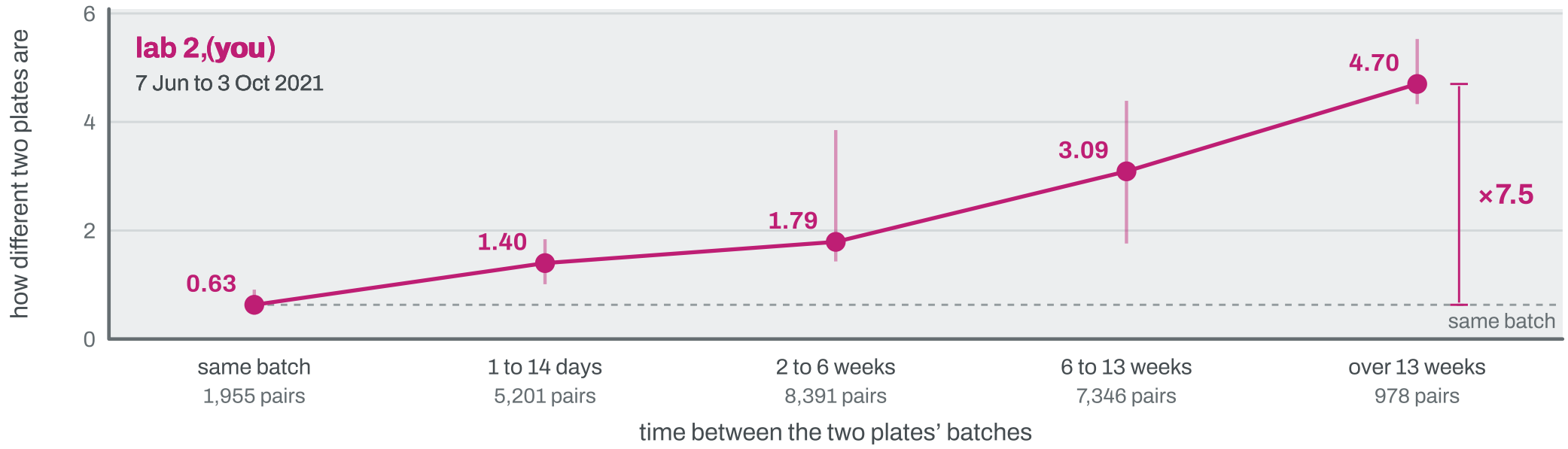
BOTH NEED A REFERENCE

Somebody else's images.
That is the map.

Fig. A5 DNA channel, 300 µm squares, JUMP cpg0016 [7], CC0, nuclei by Otsu, both ways: published [1], our tuning unpublished

Plates drift within two weeks: re-check every batch.

Two plates from one batch differ by 0.63; two weeks apart, 1.4. The jump past 13 weeks is lab 2's autumn move (A13).



IS THE CELL LINE RUINED?

This data shows when, not why.
What moved in lab 2: A12.

HOW FAST?

Within 2 weeks: ×2.2 vs same batch.
6 to 13 weeks: ×4.9.

HOW SOLID IS THE NUMBER?

One lab, 219 plates, 13 batches.
Other labs not yet measured.

dot = median pair of plates, bar = middle half of pairs, 1 = how far a typical DMSO well sits from its plate's average, JUMP [7]

STR says which line; images say where it was made.

One cell line, U2OS, in all 11 labs.
Images still name the lab: 99.9% right.

STR: 17 loci + amelogenin, ATCC [14], Datum Line: 93,228 U2OS DMSO wells, 1,872 plates, 11 labs, plates held out, JUMP [7]

Known drugs show whether the test itself worked.

DMSO wells show where a lab sits.
Lab 3: 4 batches where the known drugs did not work.

failed = under ¼ of the lab’s usual effect + unlike other labs, 1,829 plates, maps: plates C13443aW, A12069aW, JUMP layout [7]

The same drug looks less alike in another lab.

RSL3 at 10 µM from fig 4a, every well against every other.
Darker = more alike.

agreement = cosine of two wells’ effects, each well minus its own plate’s DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

One unstained image already names the lab.

Evidence: cells stained with six dyes.
Product: plain light, no dye.

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells, split and model for every row, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

It flags rarely: six of 129 batches.

8 of 11 labs never flagged.

The one way that differs flags more, not less.

Each plate checked against its own lab’s first batch, 1,871 plates, 11 labs, JUMP [7], lab 2 over time: A13

Stain brightness rose; the cell count did not.

Lab 2 (you): flagged batches against its own “normal”, its first batch (7 Jun).

ER, AGP, Mito, DNA: four stain channels, each marking different parts of the cell, each bar averages one channel’s measurements, JUMP [7]

By its own count lab 2 was fine; beside other labs, it had moved.

Lab 2 (you), 13 batches, two views.
Not a verdict: a reason to look.

distance unit: usual spread between wells on one plate, lab 2, 2021, JUMP [7], which batches: A11, what moved: A12

On the map, labs agree a little more on the same drug.

Seven drugs with a known effect, 125 batches, 11 labs. Higher is better.

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

Labs join free, and companies pay to use the map.

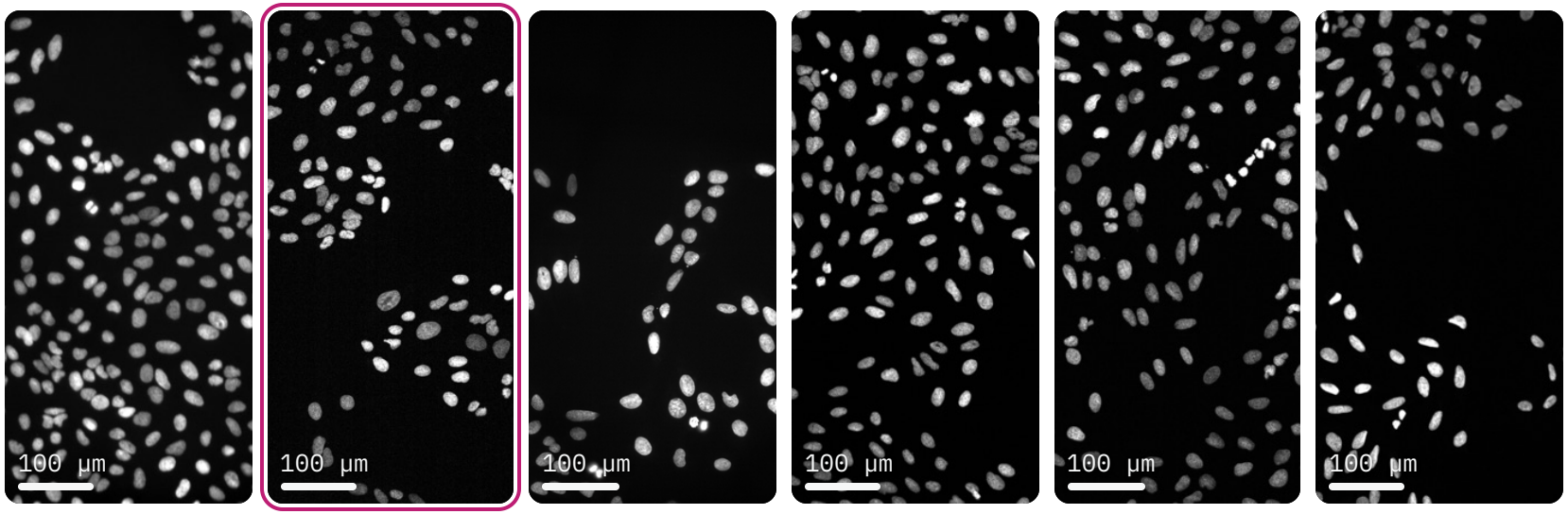
Certificate pays first.
Then the exchange: access, not ownership.

Biology's ugliest dataset, built on purpose.

4 cell lines × 2 microscope types × known drugs
+ planted faults. Every correction method scored.

TODAY, SAME CELLS, SAME PROTOCOL, NO DRUG

real wells, JUMP [7]



LAB 1	LAB 2 (YOU)	LAB 3	LAB 5	LAB 6	LAB 10
Phenix, widefield	CV8000, confocal	Phenix, widefield	CV8000, confocal	CV8000, confocal	CV8000, confocal

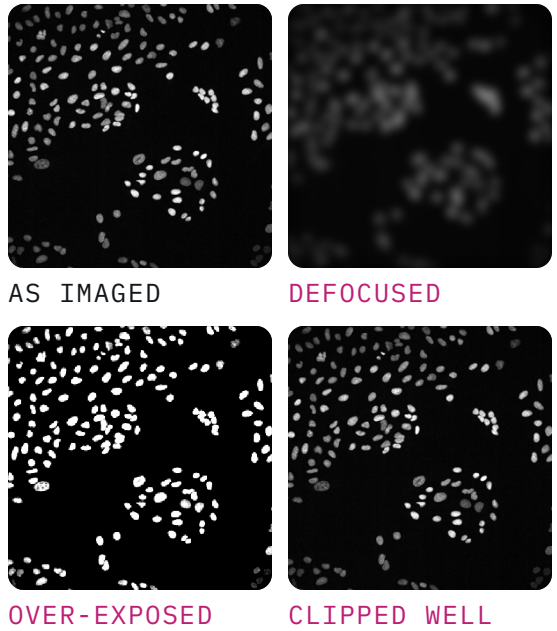
5 microscope models

8 plate layouts

13 labs [7]

PLANTED FAULTS

simulated



AS IMAGED

DEFOCUSED

OVER-EXPOSED

CLIPPED WELL

Can a model name the fault?

cell lines: U2OS, A549, HepG2, iPSC [17], microscope types: widefield + confocal

Ranked on controls, never on results.

Every lab scored on the other labs’ scale, never its own.
Lab 2 (you): 8th of 11 labs.

Lab medians, 1,871 plates, 11 labs, clean run: batches in a row never flagged (A11), JUMP [7]

Only control wells leave the lab.

No compound, no target, no result.
Nothing for legal to approve.

An image goes in; a position and a correction come out.

Results stay in the lab.
DMSO wells and the correction travel.

Correction, per batch on 200 components: $x' = C_{field}^{\frac{1}{2}} C_{batch}^{-\frac{1}{2}} (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

Every lab leaves a fingerprint, so a wrong label stands out.

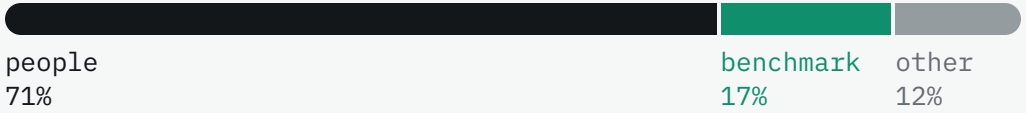
One map of DMSO wells, no results stored.
Two checks work today, two are proposed.

Integrity check uses each plate’s known-drug wells (A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

About $1.8M for eighteen months, mostly people.

$884k planned × 2 = $1.77M asked



salaries below market, leader and scientific lead paid, nothing here committed

About $84M a year, if every likely buyer joined.

Count × price.
Three streams.

prices proposed on A15, academic labs join free, they are the map, not the revenue

Cell data sells, but data from different labs does not add up.

Three public prices.
Same cells, four labs, different results.

[7] JUMP, 2023, [10] TechCrunch, 2025, [11] Recursion, 2025, [12] eWeek, 2026, [13] Wolff et al., *iScience*, 2025, full references in the README

The lab keeps its data and a share; anyone can check it.

Subscribers get access, not ownership.
No crypto token.

public sources [6], [7], [9]

Today each lab judges its own “normal”, alone.

WHY NOW

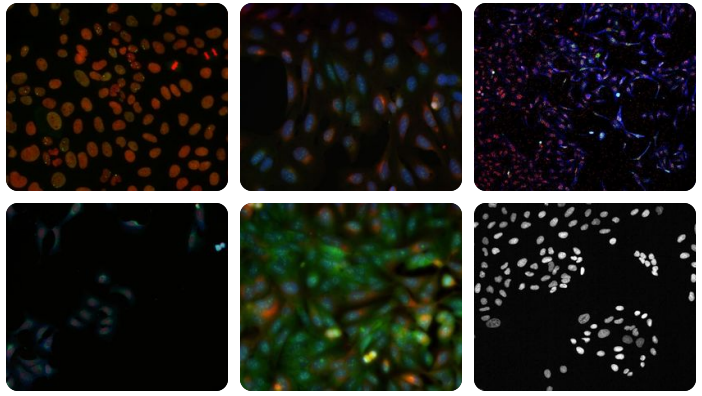
- 2023 — public cell image set [7]
- 2025 — FDA: fewer animal tests [9]
- 2026 — **the map**

$28B A YEAR, SPENT ON US LAB RESEARCH OTHERS CANNOT REPRODUCE [6]

36% ≈ $10B, cells and reagents | 64%, other causes

1
search

do my U2OS cells look “normal”

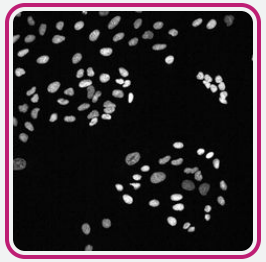


one cell line, six papers, all look different

2
ask

To: whoever ran this line last

Subject: does this look right to you?



screenshot.png

one screenshot, one opinion

3
guess

$ RUN IT ANYWAY

hundreds of thousands

of R&D dollars at risk

WAIT TO BE SURE

months

of development lost

[6] Freedman et al., *PLoS Biology*, 2015, [7] JUMP, 2023, [9] US FDA, 2025, full references in the README